

# Investor Trend Analysis Dashboard

## 1. Project Overview

### Project Title

Investor Trend Analysis Dashboard using Power BI

### Objective

The objective of this project is to analyze investor behavior, investment patterns, startup performance, and return on investment (ROI) trends using Power BI. The dashboard provides actionable insights that help stakeholders understand investment distribution, startup growth potential, investor engagement, and overall investment performance.

### Business Problem

Most businesses incur huge losses due to their lack of understanding of their company investment patterns. This use case aims to help businesses understand their investors and cash flow into their company.

---

## 2. Research Plan

### Research Questions

The following business questions were identified before starting the analysis:

1. Which industries attract the highest investment amounts?
2. Which industries generate the highest ROI?
3. Which investor types contribute the most capital?
4. Which funding stages provide the best investment returns?
5. Which countries receive the highest investment activity?
6. What percentage of investments are profitable?
7. How actively do investors participate in follow-on investments?
8. What is the level of investor involvement through board seats?
9. What data quality issues exist within the dataset?
10. How can investment trends be visualized for faster decision-making?

---

## 3. Dataset Understanding

The generated dataset consists of investor and startup-related information.

### Key Attributes

- Investor ID
- Investor Type
- Startup Name
- Industry
- Funding Stage
- Country
- Investment Amount
- Startup Valuation
- ROI Percentage
- Exit Status
- Follow-On Investment
- Board Seat Participation
- Holding Period
- Investor Satisfaction Score

### Expected Outcome

Generate business-focused KPIs and visualizations that reveal investment trends and startup performance.

# 4. Implementation Plan

## Step 0: Dataset Creation and basic excel cleaning

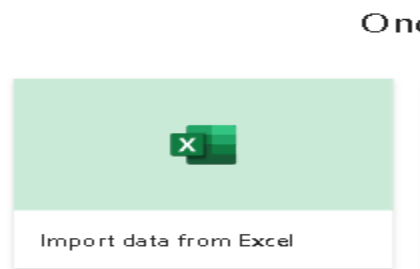
| investor_id | investor_name        | investor_type  | country   | startup_company  | industry      | funding_stage | investment_date | investment_amount_usd | startup_valuation_usd | roi |
|-------------|----------------------|----------------|-----------|------------------|---------------|---------------|-----------------|-----------------------|-----------------------|-----|
| 12825       | derek hughes         | private equity | India     | cloudmint        | e-commerce    | series b      | 29-01-2024      | 355692.82             | 3793850.32            |     |
| 10149       | craig tucker         | retail         | Canada    | neoretail        | cleantech     | series a      | 19-05-2024      | 265030.05             | 5763145.91            |     |
| 12028       | alan blake           | private equity | Singapore | healthsphere     | healthtech    | seed          | 06-02-2024      | 476222.09             | 8758436.03            |     |
| 12401       | elizabeth thomas phd | angel          | Germany   | cloudmint        | cybersecurity | ipo           | 03-08-2024      | 289036.53             | 10627292.06           |     |
| 13207       | edward beck          | private equity | USA       | quantumleap      | cleantech     | seed          | 05-02-2022      | 372526.51             | 13567510.54           |     |
| 11795       | donald long          | institutional  | Australia | cybersecurity    | cybersecurity | series b      | 12-01-2024      | 505798.99             | 3949894.14            |     |
| 12566       | collin webster       | angel          | Germany   | healthsphere     | agritech      | series a      | 30-04-2026      | 611223.97             | 13567452.09           |     |
| 12023       | jennifer berg        | retail         | Australia | visionware       | e-commerce    | series c      | 11-02-2023      | 375661.41             | 3965776.13            |     |
| 12357       | sydney coleman       | angel          | Canada    | visionware       | IoT           | series a      | 19-05-2025      | 265801.19             | 14719768.52           |     |
| 13794       | eric gray            | vc             | Germany   | smartlive        | e-commerce    | series b      | 13-10-2023      | 142568.04             | 9187325.88            |     |
| 11044       | travis smith         | retail         | USA       | urbanest         | fintech       | series c      | 04-09-2022      | 67199.67              | 10165302.53           |     |
| 13404       | alec hunter          | institutional  | India     | quantumleap      | e-commerce    | ipo           | 07-07-2023      | 667359.59             | 2346638.66            |     |
| 11149       | mrs. jill kane       | private equity | India     | technova         | fintech       | series c      | 03-04-2023      | 298758.38             | 8345537.34            |     |
| 11872       | rebecca smith        | retail         | UK        | visionware       | IoT           | series b      | 22-04-2025      | 457903.77             | 11561464.78           |     |
| 11558       | ronald horn          | retail         | India     | finedge          | e-commerce    | series b      | 18-10-2021      | 705695.7              | 12771076.13           |     |
| 13641       | brian mccann         | private equity | USA       | quantumleap      | cybersecurity | seed          | 22-06-2022      | 435069.71             | 5653156.75            |     |
| 13814       | valerie bentley      | retail         | USA       | quantumleap      | e-commerce    | ipo           | 17-06-2022      | 472845.84             | 13492264.27           |     |
| 10812       | dr. elizabeth miller | angel          | Germany   | greempulse       | fintech       | series c      | 31-07-2025      | 467131.85             | 8252943.05            |     |
| 11511       | susan jackson        | retail         | USA       | technova         | IoT           | series b      | 15-05-2025      | 280381.55             | 7449649.93            |     |
| 13960       | michael torres       | retail         | USA       | nextgen robotics | fintech       | ipo           | 17-10-2023      | 520826.34             | 8674002.45            |     |
| 13472       | randy adams          | vc             | Canada    | databridge       | agritech      | series b      | 26-06-2024      | 818107.1              | 13419615.69           |     |
| 11116       | brian thompson       | institutional  | Canada    | greempulse       | edtech        | series a      | 14-01-2024      | 669732.82             | 7922414.51            |     |
| 11840       | jennifer cobb        | vc             | Australia | visionware       | fintech       | series b      | 14-12-2025      | 313514.47             | 13013677.85           |     |
| 12981       | Nithya Sri           | private equity | Singapore | visionware       | saas          | series c      | 15-01-2024      | 586849.37             | 9411229               |     |
| 11417       | kyle turner          | retail         | India     | healthsphere     | edtech        | series b      | 15-11-2025      | 552667.12             | 5633499.31            |     |
| 11891       | tanya solis          | angel          | India     | blueorbit        | saas          | series b      | 17-07-2022      | 453892.14             | 10397255.26           |     |
| 12321       | sandra               | vc             | Canada    | finedge          | cleantech     | ipo           | 29-10-2022      | 414462.84             | 12149914.82           |     |
| 12763       | sergio basu          | private equity | Singapore | nextgen robotics | IoT           | ipo           | 18-04-2022      | 557677.04             | 11275390.16           |     |

## Data Cleaning and Preprocessing

1. **Removed duplicate records** to ensure accurate investor counts and investment calculations.
2. **Handled missing values** by replacing blank categorical fields with "Unknown" to maintain consistency in reporting.
3. **Standardized country names** by grouping variations such as *USA*, *United States*, and *Capital USA* into a single category (*United States*).
4. **Cleaned Investor IDs** by removing unnecessary prefixes and irrelevant text to maintain a uniform identifier format.
5. **Corrected data inconsistencies** across columns such as Industry, Funding Stage, and Exit Status by standardizing naming conventions.
6. **Validated and corrected data types** for numerical fields like Investment Amount, Startup Valuation, ROI Percentage, and Holding Period.
7. **Prepared the final cleaned dataset** for Power BI analysis, ensuring reliable KPI calculations, filtering, and visualization outputs.

## Step 1: Data Acquisition

The investment dataset was imported into Power BI Desktop from an Excel source file.

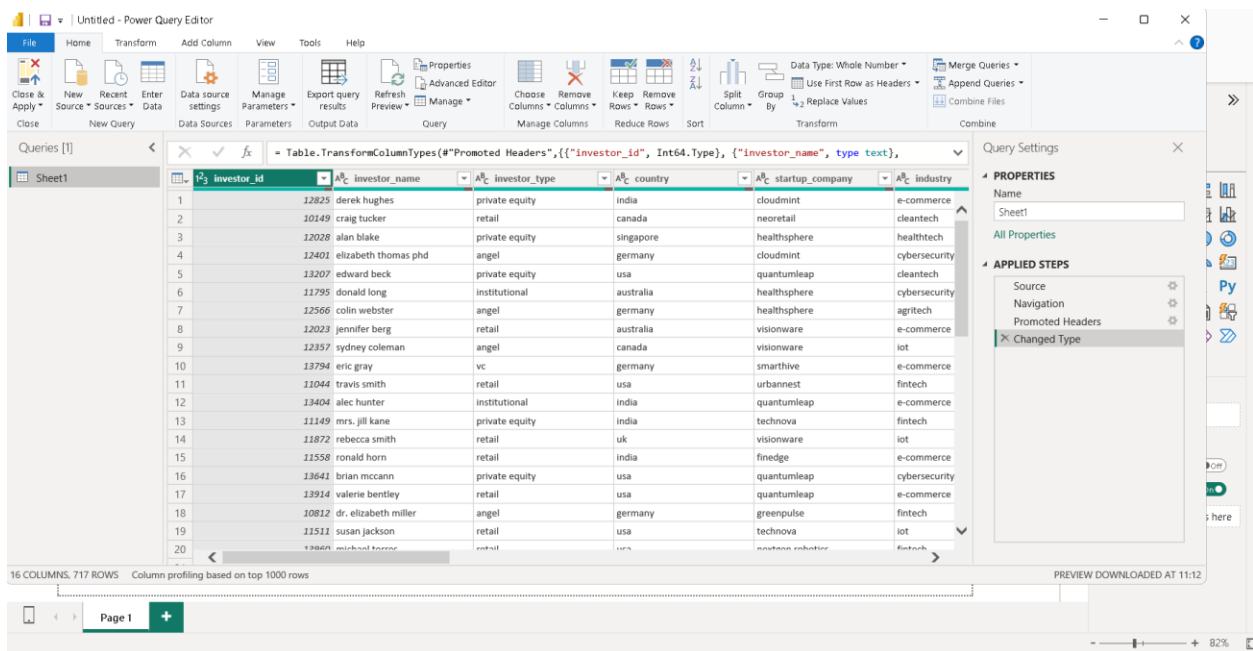


---

## Step 2: Data Profiling

The dataset was inspected to understand:

- Number of records
- Number of columns
- Missing values
- Duplicate records
- Data types



## Step 3: Data Cleaning

The following data quality procedures were performed:

### Handling Missing Values

Missing values were identified in:

- Investment Amount
- ROI
- Startup Valuation

### Duplicate Detection

Duplicate investor records were identified and analyzed.

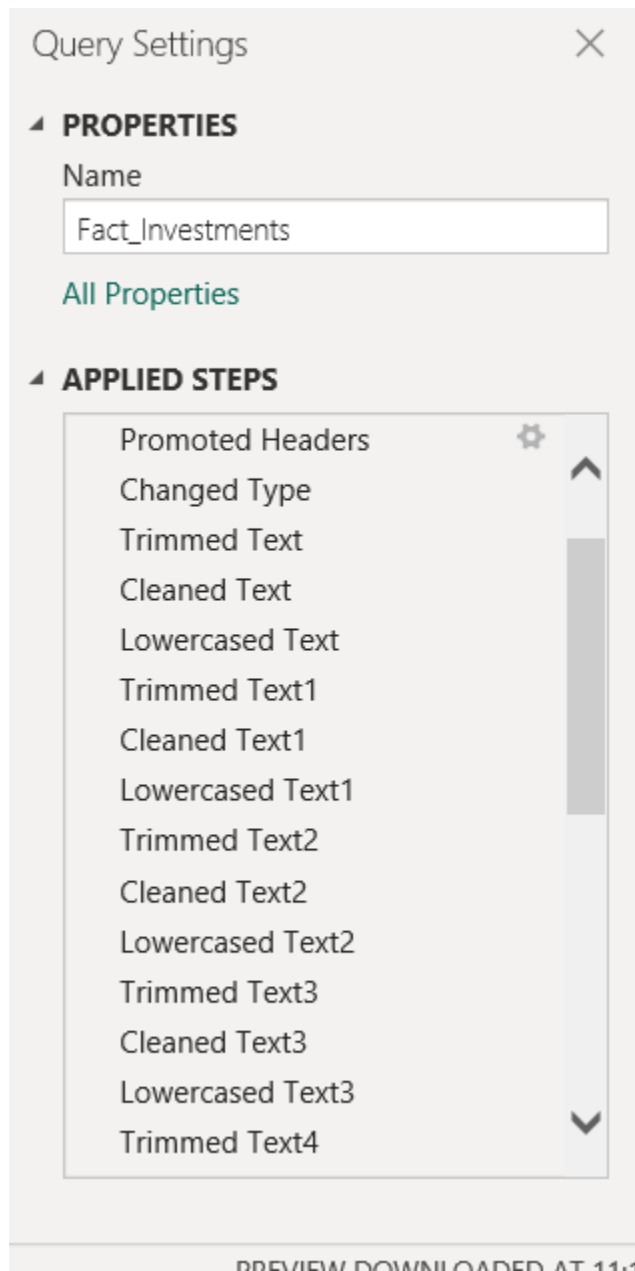
### Data Type Validation

Numerical columns were converted into appropriate data types:

- Decimal Number
- Whole Number
- Percentage

## Outlier Validation

ROI values outside expected business ranges were identified.



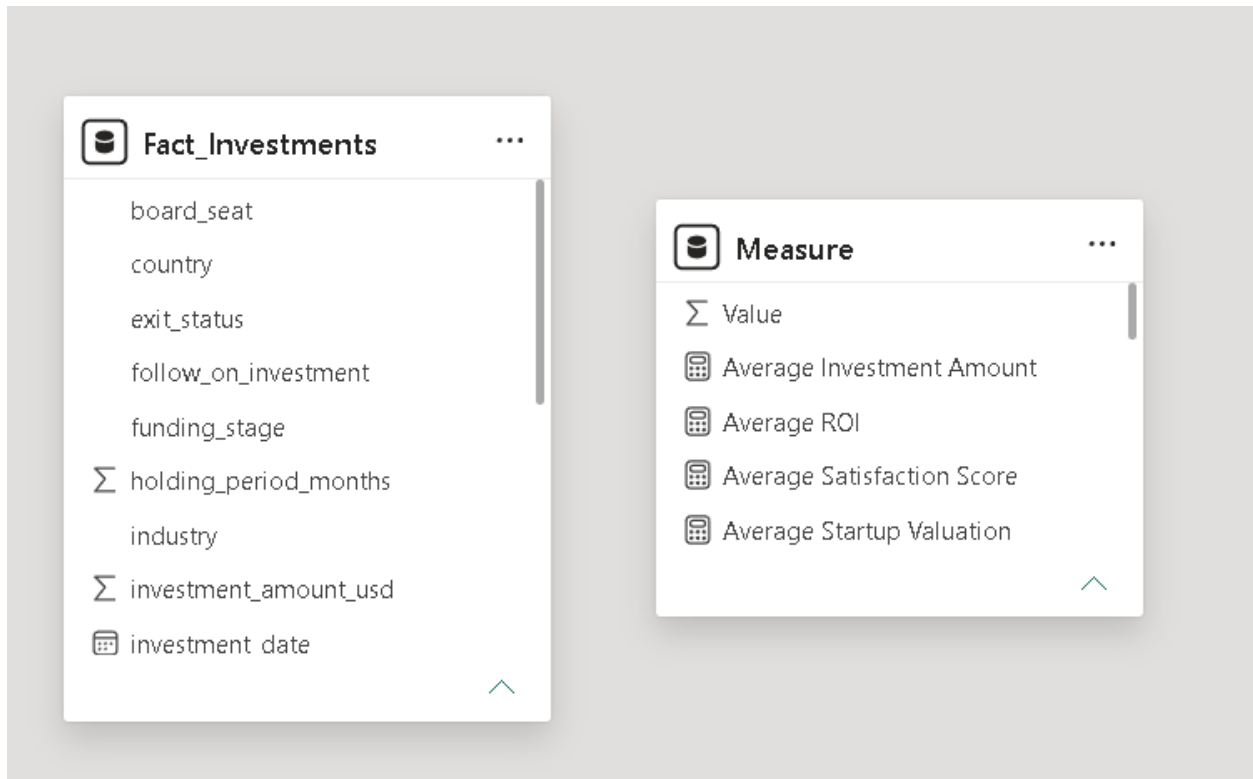
## Step 4: Data Modeling

To maintain simplicity for the internship use case, a lightweight data model was adopted.

### Tables Used

1. Cleaned\_investor\_data a.k.a Fact\_Investments (power BI)
2. Measures

No additional dimension tables were created because the dataset structure was sufficient for analysis.



## Step 5: KPI Development

The following DAX measures were created:

### Investment KPIs

- Total Investors
- Total Investment Amount
- Average Investment Amount
- Total Startup Valuation

### Performance KPIs

- Average ROI
- Highest ROI
- Lowest ROI
- Success Rate

### Investor Engagement KPIs

- Follow-On Investment Rate
- Board Seat Rate
- Average Holding Period
- Average Satisfaction Score



## Step 6: Dashboard Design

Three dashboard pages were created.

### Page 1: Executive Summary

Visuals:

- KPI Cards
- Investment Amount by Industry
- ROI by Industry
- Investor Type Distribution
- Funding Stage Distribution

### Page 2: Investor and Startup Analysis

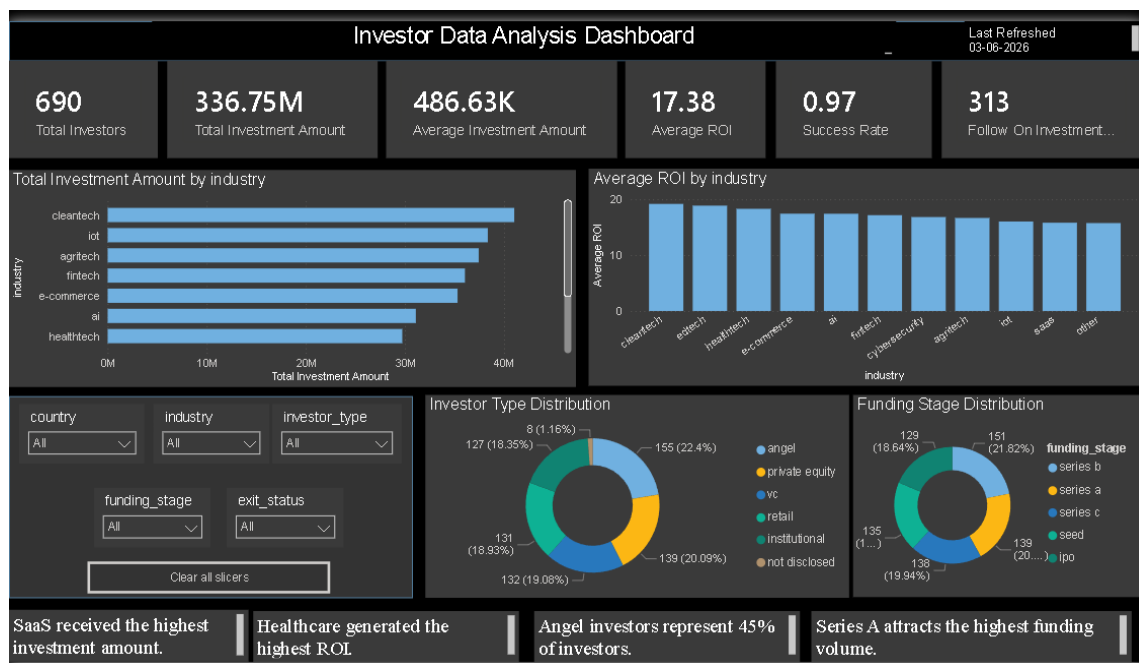
Visuals:

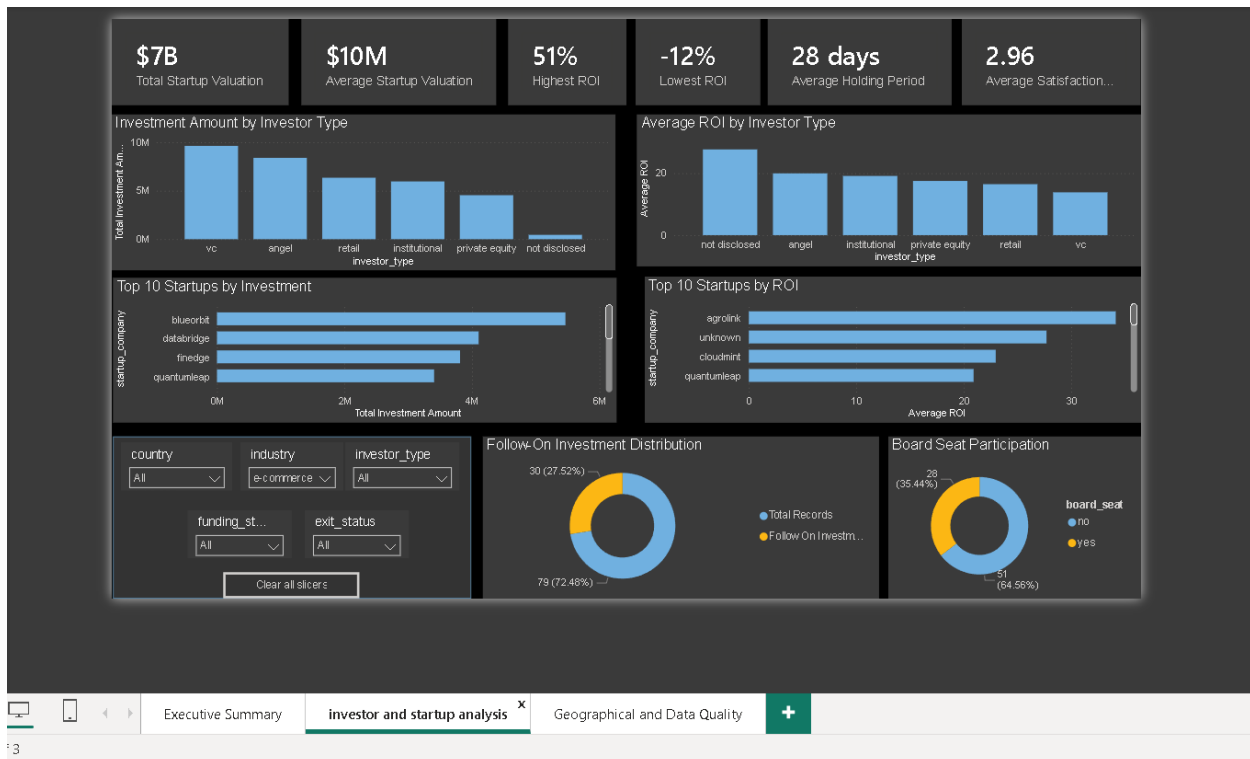
- Investment Amount by Investor Type
- ROI by Investor Type
- Top 10 Startups by Investment
- Top 10 Startups by ROI
- Board Seat Participation
- Follow-On Investment Distribution

### Page 3: Geographic and Data Quality Analysis

Visuals:

- Investment Amount by Country
- ROI by Country
- Investors by Country
- Data Quality Summary

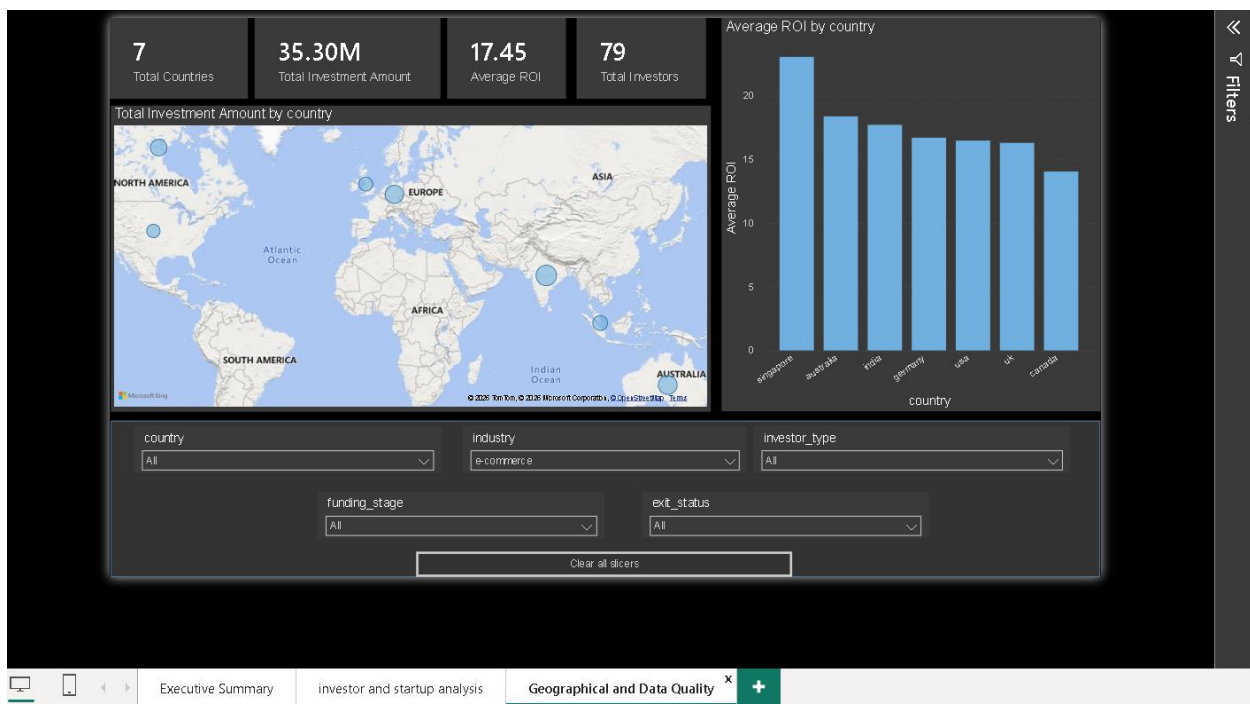




Executive Summary

investor and startup analysis

Geographical and Data Quality



Total Investment Amount by country

Average ROI by country

| Country   | Average ROI (%) |
|-----------|-----------------|
| singapore | 22              |
| australia | 18              |
| india     | 17              |
| germany   | 16              |
| usa       | 16              |
| uk        | 16              |
| canada    | 14              |

country

All

industry

e-commerce

investor\_type

All

funding\_stage

All

exit\_status

All

Clear all slicers

Executive Summary

investor and startup analysis

Geographical and Data Quality

# KDataset Summary

| Metric                        | Value          |
|-------------------------------|----------------|
| Total Records                 | 3,807          |
| Unique Investors              | 3,732          |
| Total Investment Amount       | \$1.85 Billion |
| Average Investment Amount     | \$500,919      |
| Average ROI                   | 17.46%         |
| Success Rate                  | 94.9%          |
| Follow-On Investment Rate     | 48.57%         |
| Board Seat Participation Rate | 48.12%         |

---

## Key Business Findings

### Finding 1: Strong Overall Investment Performance

The investment portfolio demonstrated positive performance with an average ROI of **17.46%**.

#### Interpretation

- The majority of investments generated returns above the initial investment.
- The portfolio maintained healthy profitability across industries and funding stages.

#### Supporting Visual

Average ROI KPI Card  
ROI by Industry Chart

---

## Finding 2: High Investment Success Rate

Approximately **94.9%** of investments generated positive returns.

### Interpretation

- The portfolio contains a very high percentage of profitable investments.
- Investment selection strategies appear effective.

### Supporting Visual

Success Rate KPI Card

---

## Finding 3: FinTech Attracted the Highest Investment Volume

Top industries by total investment amount:

| Industry   | Investment Volume |
|------------|-------------------|
| FinTech    | \$206.95 Million  |
| CleanTech  | \$196.36 Million  |
| AgriTech   | \$187.15 Million  |
| EdTech     | \$187.08 Million  |
| HealthTech | \$186.66 Million  |

### Interpretation

- Investors showed the strongest confidence in financial technology startups.
- FinTech continues to dominate investment activity.

### Supporting Visual

Investment Amount by Industry

---

# Finding 4: SaaS Generated the Highest Returns

Top industries by ROI:

| Industry      | Avg ROI |
|---------------|---------|
| SaaS          | 18.38%  |
| E-Commerce    | 17.96%  |
| Cybersecurity | 17.87%  |
| IoT           | 17.54%  |

## Interpretation

- SaaS startups delivered superior returns compared to most sectors.
- Investors seeking higher growth opportunities may favor SaaS businesses.

## Supporting Visual

Average ROI by Industry

---

# Finding 5: Angel Investors Contributed the Largest Capital

Top investor categories:

| Investor Type | Total Investment |
|---------------|------------------|
| Angel         | \$383.26 Million |
| Retail        | \$362.73 Million |
| VC            | \$358.59 Million |

Private Equity    \$358.47 Million

Interpretation

- Angel investors played a major role in startup funding.
- Early-stage investment participation remains strong.

Supporting Visual

Investor Type Distribution  
Investment Amount by Investor Type

---

Finding 6: Canada Received the Highest Investment Volume

Top countries by investment amount:

| Country   | Investment Volume |
|-----------|-------------------|
| Canada    | \$285.06 Million  |
| Germany   | \$271.90 Million  |
| Australia | \$265.08 Million  |
| India     | \$260.18 Million  |
| USA       | \$252.26 Million  |

Interpretation

- Canada emerged as the leading destination for startup investments.
- Investment activity was distributed across multiple global startup ecosystems.

Supporting Visual

Investment Amount by Country

---

## Finding 7: Follow-On Investment Activity Was High

### Result

Follow-On Investment Rate:

48.57%

### Interpretation

- Nearly half of investors continued investing in existing startups.
- This indicates confidence in startup performance and growth prospects.

### Supporting Visual

Follow-On Investment Rate KPI

Follow-On Distribution Donut Chart

---

## Finding 8: Strong Investor Engagement

### Result

Board Seat Participation:

48.12%

### Interpretation

- Almost half of investors actively participated in startup governance.
- Board representation suggests long-term strategic involvement.

### Supporting Visual

Board Seat Participation KPI

Board Seat Distribution Chart

---

## Finding 9: Series C Investments Delivered Strong Returns

Funding-stage ROI ranking:

| Funding Stage | Avg ROI |
|---------------|---------|
| Series C      | 17.73%  |
| Seed          | 17.72%  |
| Series A      | 17.61%  |
| IPO           | 17.24%  |
| Series B      | 16.98%  |

### Interpretation

- Later-stage startups provided slightly stronger returns.
- Series C investments balanced risk and profitability effectively.

### Supporting Visual

ROI by Funding Stage

---

## Finding 10: Acquired and Exited Startups Generated Better ROI

Exit status ROI:

| Exit Status | Avg ROI |
|-------------|---------|
| Exited      | 17.91%  |
| Acquired    | 17.82%  |
| Active      | 17.82%  |
| Merged      | 16.39%  |

## Interpretation

- Successful exits generally produced stronger returns.
- Acquisition events contributed positively to investor profitability.

## Supporting Visual

ROI by Exit Status

---

# Technical Learnings from the Project

You should include this section in your report.

## Data Analytics

- Data profiling and exploration
- KPI identification and business analysis
- Trend analysis
- ROI evaluation

## Power BI

- Dashboard design
- Interactive visualizations
- Slicer implementation
- KPI card development
- Report storytelling

## DAX

- Aggregate functions
- Conditional calculations
- Percentage metrics
- Business KPI creation

## Data Preparation

- Handling missing values
- Duplicate detection
- Data type validation
- Data quality assessment

